Video Transcript

Video 7: Using DASK for Parallel Processing Across Distributed Machines (04:07)

So, let's take a look at DASK running on different machines. I'm going to simulate different machines on my single machine here, but let's first, we're going to launch the dask-scheduler. So, this tells us which port it's running on. So, the scheduler is running on this port. So, what we'll do now is set up two workers. So, it's 8786. So, now we've got a dask-worker connected to the scheduler.

Let's run another dask-worker and connect to the same scheduler. So, here we are now. So, we've got two of them. Now, let's get a client here. So, we need to run Python. So, here we need to say 'from dask.distributed import Client.' And now we say 'client' equals 'Client' and we need to paste in address. And now, we've got our client connected. You see here, we've got our client talking to the scheduler. Now, we can put our program into here.

So, let's type our program. So, we need to import dask.array as da. Now, we'll set up our variables so, da.ones and we'll set up 100,000 by 100,000. Okay, and I need an extra bracket in here.

So, we've got X set up. Now, we can run our calculation. Y equals da.exp(x).sum. Now, dot compute will spread the computation now across these machines, these workers. So, this normally takes about 25 seconds, and we've got the result. I can check that, but that's correct. So, this is a very nice distributed computing environment. DASK allows us to communicate across machines, basically distribute our computation in a parallel fashion. So, let's just take down the scheduler and we'll see that the other machines. So, let's just control C and notice the other machines, notice the connection is broken and they'll eventually log out. There we go. They're all logged out. Okay, so that's an exciting environment to program in. Bye for now.

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